

DC electric field induced phase array self-assembly of Au nanoparticles

This content has been downloaded from IOPscience. Please scroll down to see the full text.

2014 Nanotechnology 25 465301

(<http://iopscience.iop.org/0957-4484/25/46/465301>)

View [the table of contents for this issue](#), or go to the [journal homepage](#) for more

Download details:

IP Address: 147.231.26.31

This content was downloaded on 02/04/2015 at 12:39

Please note that [terms and conditions apply](#).

DC electric field induced phase array self-assembly of Au nanoparticles

S Yadavali¹, R Sachan², O Dyck³ and R Kalyanaraman^{1,3}

¹Department of Chemical and Biomolecular Engineering, University of Tennessee, Knoxville, TN 37996, USA

²Material Science and Technology Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA

³Department of Material Science and Engineering, University of Tennessee, Knoxville, TN 37996, USA

E-mail: ramki@utk.edu

Received 21 June 2014, revised 8 September 2014

Accepted for publication 16 September 2014

Published 30 October 2014

Abstract

In this work we report the discovery of phase array self-assembly, a new way to spontaneously make periodic arrangements of metal nanoparticles. An initially random arrangement of gold (Au) or silver (Ag) nanoparticles on SiO₂/Si substrates was irradiated with linearly polarized (*P*) laser light in the presence of a dc electric (*E*) field applied to the insulating substrate. For *E* fields parallel to the laser polarization (*E* ∥ *P*), the resulting periodic ordering was single-crystal like with extremely low defect density and covered large macroscopic areas. The *E* field appears to be modifying the phase between radiation scattered by the individual nanoparticles thus leading to enhanced interference effects. While phase array behavior is widely known in antenna technology, this is the first evidence that it can also aid in nanoscale self-assembly. These results provide a simple way to produce periodic metal nanoparticles over large areas.

Keywords: nanoparticles, phase array, self-assembly, pulsed laser, interference, gold, electric field

(Some figures may appear in colour only in the online journal)

1. Introduction

Spatially periodic metal nanoparticle arrays are important in a wide range of applications, such as for plasmonic waveguiding [1], for nanoscale lensing [2, 3], for catalyzing the growth of ordered carbon nanotubes [4] and as Raman active substrates [5]. However, one of the challenges is to fabricate large area periodic arrays of nanostructures in a reliable and cost effective way, and this has led to intense research in nanoscale fabrication methods. Several fabrication techniques based on lithography and bottom up self-assembly routes have emerged to synthesize nanoparticles in arbitrary geometries, size, and one- or two-dimensional (1D or 2D) spatial arrangement. Lithography techniques utilizing ultraviolet (UV), e-beam or ion beam energies are all powerful ways, but are ultimately limited by their high cost, complicated mask technology, development of appropriate resist materials and most importantly, requirement of sophisticated instrumentation. Laser-induced forward transfer (LIFT) [6, 7] and a combination of focused ion beam (FIB) and thermal

annealing [8] have also been used to synthesize 1D particle arrays. The major limitations of LIFT are the difficulty in controlling the inter particle spacing in sub-micron scale and achieving large area of nanoparticle arrays, since particles are produced in a serial manner. Thermal annealing is complicated because of metal–substrate chemical interactions and metal diffusion into the substrate, which may cause impure particle arrays. Self-assembly methods like nanosphere lithography (NSL) [9] and pulsed laser induced self-organization [10–13] are easy and cost effective ways to fabricate nanostructures, but they also have their own limitations. NSL uses self-assembled monolayers of polystyrene spheres as evaporation masks, and presently, only a hexagonal geometry of nanoparticle patterns can be obtained. In laser induced self-organization techniques, multiple beam interference irradiation of thin metal films [14, 15] can produce 1D or 2D ordering of metallic nanoparticles due to thermal gradient effects. But the driving forces from the laser interference process become much weaker in materials with large thermal conductivity, such as materials like Ag and Au.

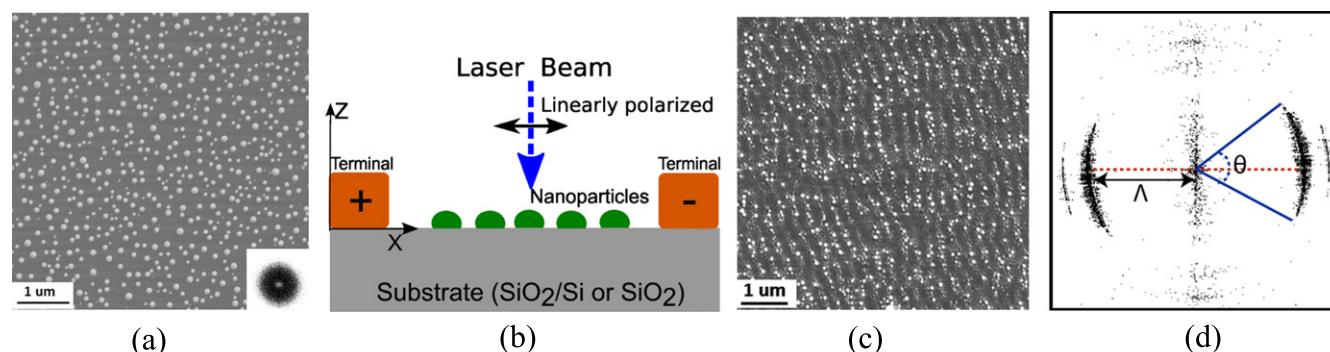


Figure 1. (a) SEM image of Au nanoparticles synthesized by pulsed laser dewetting at a laser energy density of $\rho = 60 \text{ mJ cm}^{-2}$. The inset shows the fast Fourier transform (FFT) of the image. (b) Schematic depiction of the experimental configuration of the dc electric field and the incident linearly polarized laser light. In the schematic shown, the polarization and applied field are both along the x -direction, i.e. $E \parallel P$. In the $E \perp P$, the E field was applied along the y -direction, which was perpendicular to the x - z plane shown. (c) SEM image showing domains of periodic structures misoriented with respect to each. This pattern was formed by irradiation of the nanoparticles in (a) by the laser with energy density of 90 mJ cm^{-2} and 0 V electric field. (d) Computer generated diffraction pattern (FFT) of the image in (c). The angular spread of the various domains (θ) and the spacing of the periodic pattern (Λ) was determined from such FFT patterns.

Within the context of laser interference processes, the phenomenon of laser induced periodic surface structure (LIPSS) has been widely investigated, both from a fundamental perspective as well as a way to make ordered nanostructures [16–25]. In LIPSS, a spontaneous spatially periodic pattern emerges simply under irradiation by a single coherent laser beam. The pattern is a result of an interference phenomenon between the incident light and scattered light [22, 24, 26–28]. The laser interference pattern produces periodic thermal gradients that cause mass transport and rearrangement due to thermocapillary effects [19, 29]. For this reason, LIPSS is a highly attractive phenomenon for fabrication of templates and/or nanostructures because of its extremely simple experimental implementation [30, 31]. However the patterns produced by LIPSS are often low quality, with defects like bifurcation and domain misorientation (ripples), resulting in poor ordering over large areas. In fact, the inability to eliminate defects in LIPSS patterns has limited its implementation in practical technologies.

In this work, we show a scientifically interesting and practical new way to significantly enhance the quality of the spontaneous 1D ordering arising under the irradiation of a surface by a single laser beam. When an initially near-random arrangement of metal nanoparticles on quartz or glass (SiO_2/Si) substrates was irradiated by laser light, a traditional defective LIPSS pattern resulted. However, when a dc electric field was applied to the underlying glass substrate, significant changes to the 1D ordering quality and coverage area were observed. Single-crystal like patterns over macroscopically large areas could be achieved under specific field orientation and magnitudes.

2. Experimental details

Au films of thickness $10 \pm 0.5 \text{ nm}$ were deposited in high vacuum ($\sim 1 \times 10^{-8} \text{ Torr}$) by electron beam deposition at room temperature onto transparent square shaped ($3 \times 3 \text{ mm}^2$)

quartz substrates as well as SiO_2/Si substrates consisting of optically smooth 400 nm thick thermally grown oxide layer on polished Si(100) wafers. The substrates were cleaned in methanol, acetone, and DI water. The substrates were masked on their edges by a Cu tape (double sided 3 M Cu conductive tape) prior to deposition. Following the deposition the tape was removed to leave behind the deposited film and bare substrate regions. The deposition rate used here was typically $\sim 1.5 \text{ nm min}^{-1}$ as confirmed by atomic force microscope (AFM) step height measurements of the film thickness. The surface roughness of the deposited films were measured by AFM and was established to be $0.5 \pm 0.2 \text{ nm}$ for the average root mean square roughness.

The laser irradiation was performed in two stages. In stage 1, the deposited Au or Ag films were irradiated by a 266 nm UV laser having a pulse width of 9 ns. Irradiation was performed in air ambient and at normal incidence by an unfocused laser beam of area $1 \times 1 \text{ mm}^2$ for 3000 pulses at a repetition rate of 50 Hz. The laser energy density used was $\rho = 60 \text{ mJ cm}^{-2}$. This irradiation step resulted in a near-random distribution of Au or Ag nanoparticles due to the phenomenon of spinodal dewetting, which has been described in detail previously [32]. This stage 1 nanoparticle distribution was characterized by scanning electron microscopy (SEM) and is shown in figure 1(a) for Au and in figure 2(a) for Ag. The Cu conductive tape was reapplied to the bare substrate regions prior to the stage 2 irradiation for application of the electric field. In stage 2, the Au nanoparticles were irradiated with linearly polarized (P) laser light with an energy density $\rho = 90 \text{ mJ cm}^{-2}$ in the presence of a dc electric field (E) applied to the SiO_2 layer, as shown in figure 1(b). The dc field was applied by a HP 6299A dc power supply connected to the Cu conductive tape. 3000 laser pulses were applied in this stage 2 irradiation, which was also performed in air ambient. Two different dc field orientations were investigated. In one case, the dc electric field (E) was applied parallel to the laser polarization (P), i.e. $E \parallel P$ as shown in figure 1(b), while in the second case,

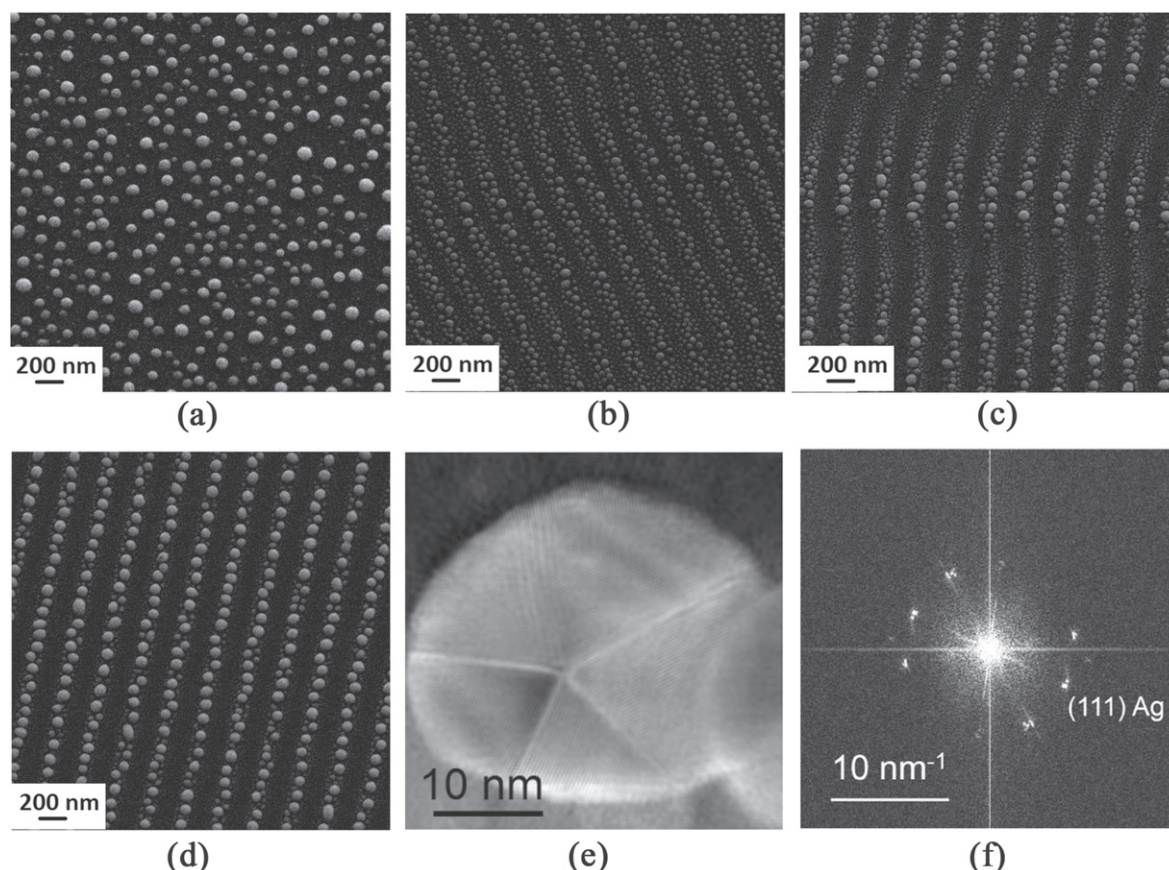


Figure 2. (a) SEM image of Ag nanoparticles synthesized by pulsed laser dewetting at $\rho = 60 \text{ mJ cm}^{-2}$. (b) SEM image showing periodic structures of Ag formed by irradiation of the nanoparticles in (a) with $\rho = 90 \text{ mJ cm}^{-2}$ and 0 V electric field. (c)–(d) SEM image of Ag periodic structures formed from (a) by irradiating with $\rho = 90 \text{ mJ cm}^{-2}$ with $E \perp P$ and $E \parallel P$ at 90 V, respectively. (e) Fourier filtered bright field cross-section TEM image of Ag nanoparticle showing polycrystalline structure. (f) FFT pattern of the nanoparticle in (e) indicated that the NP corresponds to fcc silver.

the E field was orthogonal to the laser polarization, i.e. $E \perp P$. The quality of the patterns were subsequently investigated as a function of dc field strength between 0 to 90 V, in both orientations. The Au nanoparticles on quartz, having identical characteristics of size and spacing as those prepared on the SiO_2/Si substrates, were used to measure the transmitted laser energy under application of dc E -fields. The transmitted energy was measured using an Ophir electronics PE30 A-P high power laser detector head with an Orion power meter. To determine the applicability to other materials, similar experiments were performed on silver (Ag). The stage 1 Ag nanoparticles were prepared by spinodal dewetting at laser energy density of 60 mJ cm^{-2} from 8 nm thick Ag films [33]. In stage 2, the Cu conductive tape was applied and Ag nanoparticles were irradiated with linearly polarized (P) laser light with an energy density of 90 mJ cm^{-2} . Stage 2 experiments for Ag were performed at 0 V and at 90 V using $E \parallel P$ and $E \perp P$ configurations.

High resolution surface imaging was performed using a ZEISS Merlin scanning electron microscope (SEM). AFM imaging was performed using Nanonics MultiView 1000 to analyze the topography and quality of the resulting patterns. Zeiss Libra 200 MC transmission electron microscope (TEM)

operating at 200 keV was used to determine elemental purity and crystallinity of the Au and Ag nanoparticles by performing high-angle annular dark field (HAADF) and bright field (BF) imaging, selected area diffraction (SAD) and electron energy-loss spectroscopy (EELS) studies. The cross-section TEM sample was prepared using a Zeiss Auriga SEM-FIB. A platinum–carbon (Pt–C) protective layer was first deposited over the nanoparticles made on SiO_2/Si substrates to protect them from the FIB beam [34]. A lamella was then cut from the bulk specimen with a 30 kV and 2 nA beam and lifted out *in situ*. A final polish was performed at a decreased beam current of 20 pA to minimize beam damage.

3. Results

Figure 1(a) shows the short range ordered Au nanoparticles observed following the stage 1 laser irradiation on SiO_2/Si . The pulsed laser dewetting process [32] resulted in nanoparticles with a short-range spatial order, as evident from the ring-like characteristic of the computer generated diffraction pattern (fast Fourier transform (FFT) [35]) shown in the inset of figure 1(a). The short range ordered Au patterns were then

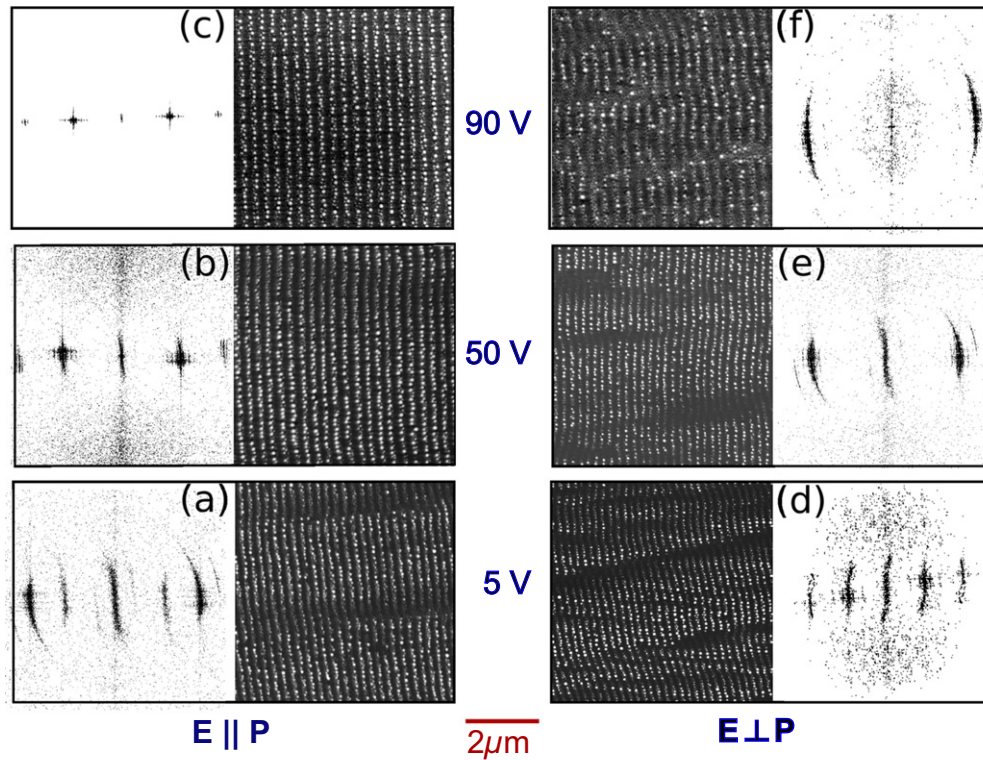


Figure 3. Result of irradiation of the nanoparticle state in figure 1(a) at laser energy density $\rho = 90 \text{ mJ cm}^{-2}$ in presence of dc electric fields of various voltages. (a)–(c) E field parallel to polarization ($E \parallel P$) at 5, 50, 90 V, respectively. The corresponding FFT's are shown besides each SEM image. (d)–(f) E field perpendicular to polarization ($E \perp P$) at 5, 50, 90 V, respectively. The corresponding FFT's are shown besides each SEM image. All images have the same scale bar of $2 \mu\text{m}$, as shown.

irradiated with linearly polarized light in presence of a dc electric field applied to the substrate, as shown by the schematic figure 1(b). Figure 1(c) show the LIPSS pattern formed when the nanoparticle array of figure 1(a) is irradiated with the higher laser energy density of $E = 90 \text{ mJ cm}^{-2}$, without any dc field. While a 1D pattern is evident, there are a large number of domains with slight misorientation in the direction of the 1D pattern. This rippled or wavy structure is reminiscent of the LIPSS patterns observed upon irradiation of bulk surfaces like Si, Ge, and Al by many authors [16, 17]. The extent of the disorder could be quantitatively represented by the angular spread in the FFT pattern, as shown in figure 1(d). This FFT pattern was also used to estimate the spacing Λ of the periodic features.

In figures 3(a)–(c), the patterns resulting from the $E \parallel P$ case for irradiation at 5, 50, and 90 V is shown. The angular spread in the orientation decreases substantially with increasing voltage, as evident from the SEM and the FFT patterns (shown beside the SEMs). In fact, a single-crystal like spot FFT pattern was evident at 90 V, signifying a very high quality of the 1D ordering.

In figures 3(d)–(f), the patterns resulting from the $E \perp P$ case for irradiation at 5, 50, and 90 V is shown. Once again, an angular spread in the 1D orientations is evident from the SEM as well as from the arc-like nature of the FFT patterns shown beside the SEMs. While the patterns again improved with increasing voltage, it started deteriorating after 50 V.

To quantitatively compare the quality of the pattern, we measured the angular spread of the FFT arcs for the various field orientations and magnitudes, and this is shown in figure 4(a). The angle was measured from the FFT pattern, as shown in figure 1(d) using a built-in angle measuring tool in the open source ImageJ [35] software. From this comparison, two important features could be gleaned. First, the quality improved significantly with increasing voltage for the $E \parallel P$ case (closed symbols). This is clearly evident from the decrease in angular spread from 66° (at 0 V) to 8° at 90 V. Secondly, the $E \perp P$ case (open symbols) was very different from $E \parallel P$. While the angular spread decreased initially with increasing voltage, the disorder increased again beyond 50 V. We also observed that changing the voltage also changed the areal coverage of the pattern under the laser irradiation zone. In figures 4(b) and (c) we show representative large area SEM images comparing the 90 V cases for the $E \parallel P$ and $E \perp P$, respectively. Clearly, it is visually apparent that $E \parallel P$ case produces a very high quality 1D ordered pattern over very large areas as compared to the $E \perp P$ case. In figure 4(d), the area of coverage of patterns observed in the $E \parallel P$ case (closed symbol) increased with increasing voltage and was orders of magnitude larger than the 0 V case. On the other hand, for the $E \perp P$ case, the area coverage initially increased with increasing voltage, but subsequently dropped and was eventually of the same order of magnitude as the 0 V case. Even at its highest coverage at 50 V, the total area was more than an order of magnitude less than the final areal coverage achieved

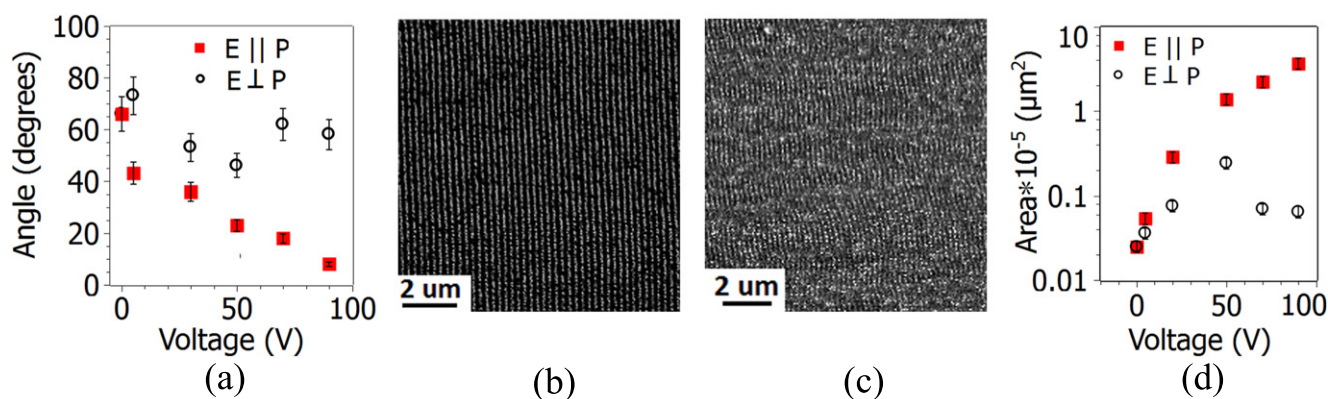


Figure 4. (a) Angular spread of the FFT arcs as a function of the E field voltage. Closed symbols correspond to $E \parallel P$ while open symbols correspond to $E \perp P$. (b) Large area SEM image of the Au patterns from $E \parallel P$ at 90 V. (c) Large area SEM image of Au patterns from $E \perp P$ setup at 90 V. (d) Area of coverage of the patterns on the sample surface as a function of applied voltage.

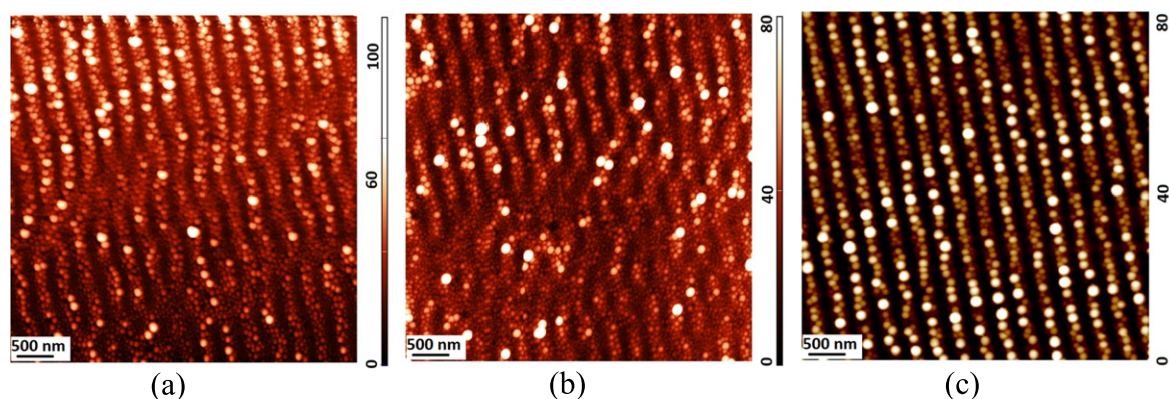


Figure 5. AFM topography images of Au periodic structures formed at laser energy density of 90 mJ cm^{-2} for various polarization and E -field cases. (a) 0 V. (b) $E \perp P$ at 90 V. (c) $E \parallel P$ at 90 V. The color bar is in units of nm.

by the $E \parallel P$ case. Similarly, the second stage irradiation of silver nanoparticles showed highest quality of patterns observed for $E \parallel P$, as shown in SEM image figure 2(d), than at 0 V, figure 2(b), or at 90 V in $E \perp P$ configuration, shown in figure 2(c).

Figure 5 shows the results of AFM topography scans from the various cases investigated following stage 2 processing of Au at 90 mJ cm^{-2} . Figure 5(a) corresponds to the nanoparticles formed at 0 V, figure 5(b) corresponds to the particles formed with $E \perp P$ at 90 V and figure 5(c) corresponds to the Au patterns formed from $E \parallel P$ at 90 V. The general observation from the AFM studies is consistent with that of the SEM in that there is very orderly arrangement of the Au NPs for the $E \parallel P$ case at 90 V, while the $E \perp P$ and 0 V show patterns with significantly lower spatial ordering of the nanoparticles.

Since the NPs have undergone high energy laser irradiation during the processing, it was important to perform a nanostructural characterization to determine the state of the Au nanoparticles. Figure 6 shows the pertinent TEM analysis of Au nanoparticles from high quality patterns obtained in $E \parallel P$ at 90 V. Figures 6(a) and (c) show a cross-section view of the Au nanoparticles on the SiO_2/Si substrate. The TEM view corresponds to looking along the rows of the

periodically arranged NPs. The overlayer on the Au NPs corresponds to the amorphous Pt-C layer deposited for the FIB sample preparation process. Figure 6(b) shows the SAD from the cross-section view of figure 6(a). The ring pattern in figure 6(b) arises from the polycrystalline Au NPs containing randomly oriented grains within each NP. Marked on the diffraction pattern are the rings corresponding to the (220), (200) and (111) planes which correspond to lattice planar spacing of 0.15 ± 0.005 , 0.21 ± 0.005 and $0.24 \pm 0.005 \text{ nm}$ respectively and is within the range for values corresponding to fcc Au. To assess if the Au NPs interact with the underlying SiO_2 substrate layer, HAADF and EELS TEM analysis was performed. The HAADF image of a typical Au NP is shown in figure 6(c), and it was evident that the interface between the NP and the underlying substrate was distinct, i.e. no burrowing or embedding into the substrate was taking place. We also confirmed that there was no interdiffusion of the substrate components (Si and O) into the Au NP by EELS maps, as shown in figure 6(d) for Si and figure 6(e) for O.

We performed similar SEM and TEM investigations of Ag nanoparticles prepared by the E -field process. Figure 2(a) corresponds to the stage 1 Ag prepared using a laser energy density of 60 mJ cm^{-2} (0 V) while figure 2(b) corresponds to

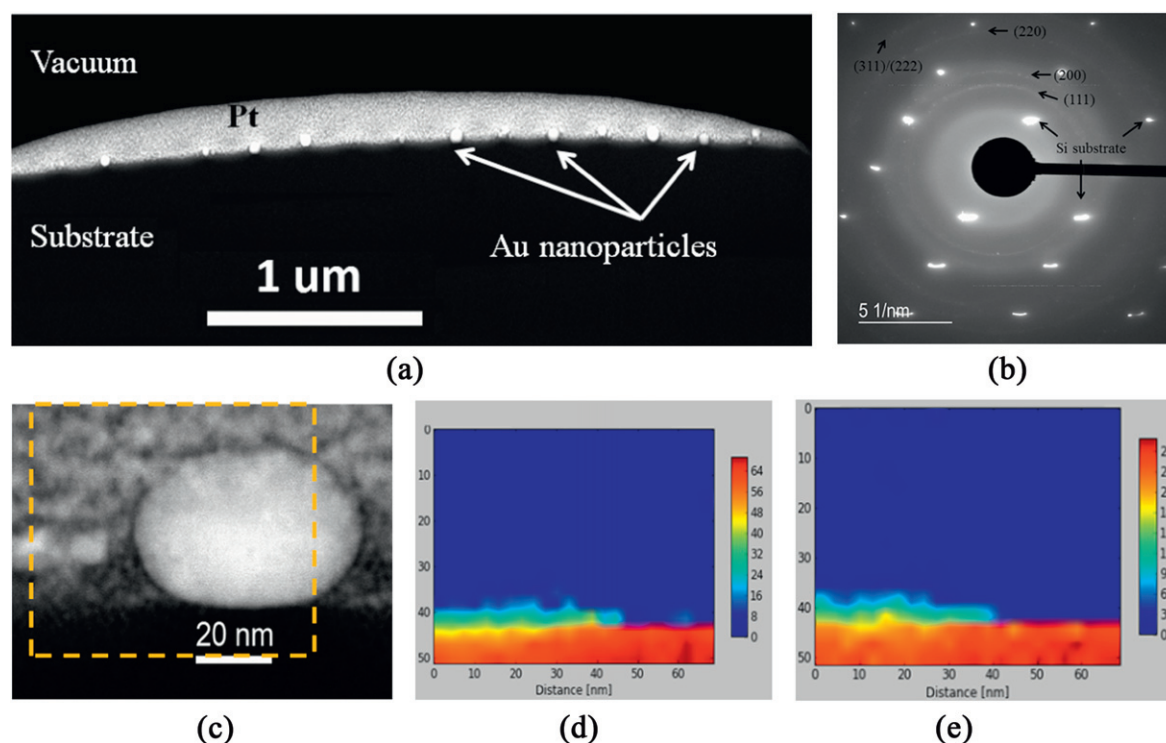


Figure 6. Cross-section TEM studies of Au nanoparticles on SiO₂/Si substrate made using $\rho = 90 \text{ mJ cm}^{-2}$. (a) Bright field view of the sample processed at 90 V for $E \parallel P$. The TEM image was obtained in a direction nearly parallel to the linear arrangement of particles in the periodic rows. The amorphous Pt + C overlayer required for the FIB processing is also evident. (b) SAD pattern from the sample shown in (a). The diffraction rings show that the Au nanoparticles are polycrystalline in nature. The bright spots correspond to the Si substrate underlying the SiO₂ layer. (c) A typical HAADF image from a single Au nanoparticle of (a) shows a well defined interface between the NP and the underlying SiO₂ layer. (d)–(e) TEM-EELS map of the regions marked in (c) for oxygen (O) (figure (d)) and silicon (Si) (figure (e)) showed that neither element was present inside the Au nanoparticles.

stage 2 Ag prepared at 90 mJ cm^{-2} and 0 V. Figures 2(c) and (d) correspond to the stage 2 Ag prepared at 90 V for the $E \perp P$ and $E \parallel P$ cases, respectively. In figure 2(e) a fourier filtered BF high resolution TEM image of a typical Ag NP from the $E \parallel P$ (90 V) case is shown. The lattice fringes are evident, as well as the presence of several grains, indicating that the NPs were polycrystalline in nature. To confirm that the synthesized Ag NPs were not contaminated by the substrate following the laser heating, the diffraction pattern from the NP was analyzed, as shown in figure 2(f). The image shows the diffraction spots corresponding to (111) lattice plane and gives a lattice spacing of $0.24 \pm 0.005 \text{ nm}$ for the NP which is consistent with the value for Ag. These results suggested that the NPs generally were pure Au and Ag metal following the irradiation at laser energy densities of 90 mJ cm^{-2} . These TEM results were also consistent with earlier results on bimetallic NPs systems synthesized on various substrates by metal dewetting under laser irradiation [36, 37].

4. Discussion

The application of a dc electric field during the process of pattern formation has been studied almost exclusively for polymer thin film systems. Applying an external electric field to an initially continuous polymer film can give rise to

modulations in surface tension, and charge density, leading to novel instabilities that can overcome the limitations encountered in conventional self-assembly methods. Electric fields can also generate electrostatic stress at the polymer and air interface due to a mismatch in their dielectric constant, and this can lead to melt flows [38]. This phenomenon has been coined as electrohydrodynamic instability. By spatially varying the electric field, dynamic patterns with 1D and 2D symmetry as well as square, rectangular, and ring-like arrangements can be generated [39]. However, similar such studies of electric field induced instabilities and patterning of metals has not been explored so far. Furthermore, such E -field effects in the context of LIPSS phenomenon has also not been explored previously.

The mechanism of the formation of periodic structures under LIPSS has been well documented. As mentioned briefly in the introduction, the thermal gradients due to the laser interference patterns are responsible for the rearrangement of matter via thermocapillary effects. A more quantitative theoretical approach to the 1D array formation under LIPSS requires solving the electrodynamic, heat, and kinetic equations [40]. In this manuscript, we are primarily interested in suggesting a possible explanation of the role of the dc E field on the LIPSS effect. Therefore, rather than providing a quantitative explanation for this phenomenon, which will be the topic of a detailed presentation in the future, we

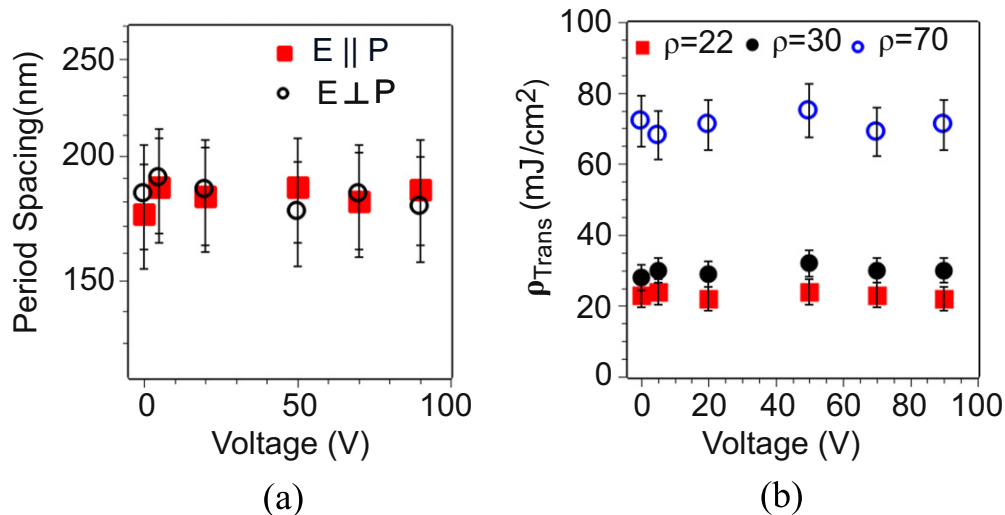


Figure 7. (a) The experimentally measured length scale of the periodic pattern as a function of voltage for the two orientations. (b) Transmitted energy of Au nanoparticles synthesized on transparent quartz substrates for different incident laser energy densities for $E \parallel P$ ($E \perp P$ gave identical results and are not shown).

hypothesize that the dc E field influences the phase between the scattered radiation from the individual nanoparticles (dipoles), analogous to the phase array behavior seen in dipole antenna radiation [41, 42]. The remainder of our discussion is focused on building a case for this hypothesis.

First, we show that the periodic patterns are consistent with an interference phenomenon between the incident radiation and scattered light. From well-known LIPPS theory [40], the periodic spacing can be expressed as $\Lambda = \lambda/(n \pm \sin \theta)$, where λ is the wavelength of laser light (266 nm), n is the refractive index of the medium producing the scattering, and θ is the incident angle measured from the normal, and in our situation is 0° . Since the scattering process is occurring in the metal + substrate region, the effective refractive index will range between the value of pure Au ($n = 1.57$ at 266 nm) and SiO_2 ($n = 1.49$ nm at 266 nm) [43]. This suggests that the observed spacing should lie between ~ 169 to 179 nm. In figure 7(a) the experimentally measured value of Λ for the two orientations and various voltages was found to be unchanged within the experimental precision and was determined to be 182 ± 12 nm, quite close to the theoretical values predicted for LIPPS.

The second important evidence concerns the behavior of the periodic pattern as a function of voltage. From figures 1(c) and (d) one can claim that the arc-like behavior of the FFT pattern is a result of small misorientations between various domains of 1D order. For the $E \parallel P$ case, increasing the voltage has the effect of aligning the various domains, hence changing the arc to a spot-like FFT pattern, as seen from figures 3(a)–(c). The situation for $E \perp P$ is similar in that, again, the effect of voltage is to rotate the domains and modify the angular spread in the FFT arcs. However, unlike the $E \parallel P$ case, a spot-like pattern is not achieved, and, in fact, at the higher voltages the misorientation begins to increase, as evident from figures 3(b)–(e). A simple explanation for the improved ordering in the $E \parallel P$ case is an improvement in the

interference quality as measured by the contrast function $CF = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$, where $I_{\max}$ and $I_{\min}$ represent the maximum and minimum in the interference intensity. If one considers as analogy a two-beam interference between linearly-polarized plane waves defined by $E(r, t) = E_i \cos(k_i r - \omega t + \phi_i)$, where E_i is the amplitude, ω is the frequency, k_i is the wave vector, and ϕ_i is the phase, then the time averaged irradiance at any point on the surface is given by $I_T(r) = I_1 + I_2 + I_{12}$, with $I_i = \langle E_i^2 \rangle_t$ being the time averaged intensity of the individual components and the interference term I_{12} given by $I_{12} = \langle \vec{E}_1 \cdot \vec{E}_2 \rangle_t = 2\sqrt{I_1 I_2} \cos \delta$, where $\delta = k_1 r - k_2 r + \phi_i - \phi_j$ represents the phase difference that is ultimately responsible for the periodic modulation of the intensity. The two extreme conditions of interference are those of total constructive interference with $\delta = \pm 2m\pi$, and total destructive interference with $\delta = \pm (m + 1/2)\pi$, where m is an integer. For this two-beam situation, the contrast function becomes $CF = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$, implying that the only way to improve the interference quality is to either increase the magnitude of the interference function $\langle \vec{E}_1 \cdot \vec{E}_2 \rangle_t$ and/or decrease the magnitude of the denominator term.

One can recognize that the denominator of the contrast function is simply the sum of the total intensity in the incident light and scattered light. Since the incident laser light intensity is fixed, the only possible way to change this function is to modify the amount of light scattered by the nanoparticle + substrate system. To measure if such an effect can occur under the E field, the total energy transmitted by the nanoparticle system, similar to that shown in figure 1(a), but prepared on a transparent quartz substrate, was measured for various normal incidence laser energy densities between 25 and 70 mJ cm $^{-2}$. The result for the $E \parallel P$ is shown in figure 7(b) and it was evident that no change in transmitted energy occurred under the E -field, indicating that the amount of scattered energy was not changed by the E field. Since the

results were identical for the $E \perp P$ case, they are not shown. This result implied that the only way to improve the CF was to increase the magnitude of the function $\langle \vec{E}_1, \vec{E}_2 \rangle_t$.

Recognizing the nature of a dot product, the increase in $\langle \vec{E}_1, \vec{E}_2 \rangle_t$ can come by a better alignment of the electric field of the incident radiation with the scattered radiation. Since the incident electric field orientation is fixed by the laser polarization, the dc electric field must be influencing the orientation or magnitude of the scattered electric field along the polarization direction. This phenomenon can be explained by the principle of pattern multiplication used in phased array dipole antennas [41, 42]. The total scattered electric field vector will be a sum over the scattered field from each nanoparticle (dipole) on the surface. Assuming a collection of identical dipole radiators, the total electric field can be expressed as $E_2 = E_o f(\phi) AF(\phi, \Lambda, k_1, \beta)$, where E_o is the scattered electric field magnitude of each dipole (assumed identical here), $f(\phi)$ is a function representing the orientation of the scattered field, and AF is the array factor that represents the magnitude of pattern multiplication. The AF depends on ϕ , k_1 and Λ , but also depends on a phase difference β . In antenna design and operation, the magnitude of β can be adjusted such that the electric field or scattered radiation is maximized in a desired direction—a principle also used large area telescopes and radar systems. We suggest here, based on the experimental evidence provided above, that the role of the dc E field is to modify β , the phase difference between the individual nanoparticle scatterers, thus modifying the interference CF. In the case of $E \parallel P$, increasing voltage has the effect of increasing the scattered field along the polarization direction thus leading to single-crystal like ordering in one dimension. On the other hand, the $E \perp P$ seems to have the effect of aligning the scattered field away from the incident polarization, this producing a poorer pattern quality. Our future work will focus on the fundamental reasons underlying the change in β with dc E -field.

5. Conclusions

In conclusion we have discovered a new phase array self-assembling phenomenon in which a dc electric field applied to an insulating substrate can control the periodic pattern formation of metallic nanoparticles under polarized laser irradiation. A near-random distribution of Au (or Ag) nanoparticles can spontaneously assemble into a 1D periodic arrangement with extremely low misorientation defects, producing a single-crystal-like spot FFT pattern. The underlying role of the electric field appears to be to change the phase between the radiation scattered by the individual nanoparticles. While this phase array effect is widely used in radar and antenna applications, it is the first demonstration of this concept in the nanoscale, and especially to self-assemble materials. This simple approach could be very useful towards fabricating metal nanoparticle arrays over large areas with a high degree of 1D periodic arrangement.

Acknowledgments

SY and RK acknowledge support by the NSF through EAGER grant CBET-1349507 and grant CBET-1402962. For the characterization techniques used in the present study, the authors thank the Sustainable Energy Education and Research Center (SEERC) and the Joint Institute of Advanced Materials (JIAM) at The University of Tennessee, Knoxville (UTK), and CNMS2013-284 at the Center for Nanophase Materials Science, which is sponsored at ORNL by the Scientific User Facilities Division, Office of Basic Energy Sciences, US Department of Energy. RK also acknowledges useful discussions pertaining to the TEM studies with Prof Gerd Duscher.

References

- [1] Maier S A, Kik P G, Atwater H A, Meltzer S, Harel E, Koel B E and Requicha A A 2003 Local detection of electromagnetic energy transport below the diffraction limit in metal nanoparticle plasmon waveguides *Nat. Mater.* **2** 229–32
- [2] Dai J, Čajko F, Tsukerman I and Stockman M I 2008 Electrodynamical effects in plasmonic nanolenses *Phys. Rev. B* **77** 115419
- [3] Li K, Stockman M I and Bergman D J 2003 Self-similar chain of metal nanospheres as an efficient nanolens *Phys. Rev. Lett.* **91** 227402
- [4] Fan S, Chapline M G, Franklin N R, Tomblor T W, Cassell A M and Dai H 1999 Self-oriented regular arrays of carbon nanotubes and their field emission properties *Science* **283** 512–4
- [5] Duan G, Cai W, Luo Y, Li Y and Lei Y 2006 Hierarchical surface rough ordered au particle arrays and their surface enhanced raman scattering *Appl. Phys. Lett.* **89** 181918
- [6] Willis D A and Grosu V 2005 Microdroplet deposition by laser-induced forward transfer *Appl. Phys. Lett.* **86** 244103
- [7] Serra P, Fernández-Pradas J M, Colina M, Duocastella M, Dominguez J and Morenza J L 2006 Laser-induced forward transfer: a direct-writing technique for biosensors preparation *J. Laser Micro/Nanoeng.* **1** 236–42
- [8] Lin C, Jiang L, Chai Y, Xiao H, Chen S and Tsai H 2010 A method to fabricate 2d nanoparticle arrays *Appl. Phys. A* **98** 855–60
- [9] Haynes C L and Van Duyne R P 2001 Nanosphere lithography: a versatile nanofabrication tool for studies of size-dependent nanoparticle optics *J. Phys. Chem. B* **105** 5599–611
- [10] Sachan R, Yadavali S, Shirato N, Krishna H, Ramos V, Duscher G, Pennycook S J, Gangopadhyay A, Garcia H and Kalyanaraman R 2012 Self-organized bimetallic ag-co nanoparticles with tunable localized surface plasmons showing high environmental stability and sensitivity *Nanotechnology* **23** 275604
- [11] Yadavali S and Kalyanaraman R 2014 Nanomaterials synthesis by a novel phenomenon: the nanoscale rayleigh-taylor instability *AIP Adv.* **4** 047116
- [12] Yadavali S, Krishna H and Kalyanaraman R 2012 Morphology transitions in bilayer spinodal dewetting systems *Phys. Rev. B* **85** 235446
- [13] Sachan R, Ramos V, Malasi A, Yadavali S, Bartley B, Garcia H, Duscher G and Kalyanaraman R 2013 Oxidation-resistant silver nanostructures for ultrastable plasmonic applications *Adv. Mater.* **25** 2045–50

- [14] Favazza C, Trice J, Krishna H, Kalyanaraman R and Sureshkumar R 2006 Laser-induced short-and long-range orderings of co nanoparticles on sio 2 *Appl. Phys. Lett.* **88** 153118
- [15] Favazza C, Trice J, Kalyanaraman R and Sureshkumar R 2007 Self-organized metal nanostructures through laser-interference driven thermocapillary convection *Appl. Phys. Lett.* **91** 043105
- [16] Young J F, Preston J, Van Driel H and Sipe J 1983 Laser-induced periodic surface structure: II. Experiments on Ge, Si, Al, and brass *Phys. Rev. B* **27** 1155
- [17] Birnbaum M 2004 Semiconductor surface damage produced by ruby lasers *J. Appl. Phys.* **36** 3688–9
- [18] Young J F, Sipe J E and van Driel H M 1984 Laser-induced periodic surface structure: III. Fluence regimes, the role of feedback, and details of the induced topography in germanium *Phys. Rev. B* **30** 2001–15
- [19] Pedraza A, Guan Y, Fowlkes J and Smith D 2004 Nanostructures produced by ultraviolet laser irradiation of silicon: I. Rippled structures *J. Vac. Sci. Technol. B* **22** 2823–35
- [20] Bonse J, Rosenfeld A and Krüger J 2009 On the role of surface plasmon polaritons in the formation of laser-induced periodic surface structures upon irradiation of silicon by femtosecond-laser pulses *J. Appl. Phys.* **106** 104910
- [21] Colombier J-P et al 2012 Plasmonic and hydrodynamic effects in ultrafast laser-induced periodic surface structures on metals *J. Laser Micro/Nanoeng.* **7** 362–8
- [22] Miyaji G, Miyazaki K, Zhang K, Yoshifuji T and Fujita J 2012 Mechanism of femtosecond-laser-induced periodic nanostructure formation on crystalline silicon surface immersed in water *Opt. Express* **20** 14848–56
- [23] Murphy R D, Torralva B, Adams D P and Yalisove S M 2014 Polarization dependent formation of femtosecond laser-induced periodic surface structures near stepped features *Appl. Phys. Lett.* **104** 231117
- [24] Sipe J, Young J F, Preston J and van Driel H 1983 Laser-induced periodic surface structure: I. Theory *Phys. Rev. B* **27** 1141
- [25] Tan B and Venkatakrishnan K 2006 A femtosecond laser-induced periodical surface structure on crystalline silicon *J. Micromech. Microeng.* **16** 1080
- [26] Scorticati D et al 2012 Ultra-short-pulsed laser-machined nanogratings of laser-induced periodic surface structures on thin molybdenum layers *J. Nanophotonics* **6** 063528
- [27] Skolski J et al 2012 Laser-induced periodic surface structures: fingerprints of light localization *Phys. Rev. B* **85** 075320
- [28] Yao J-W, Zhang C-Y, Liu H-Y, Dai Q-F, Wu L-J, Lan S, Gopal A V, Trofimov V A and Lysak T M 2012 High spatial frequency periodic structures induced on metal surface by femtosecond laser pulses *Opt. Express* **20** 905–11
- [29] Longstreth-Spoor L, Trice J, Garcia H, Zhang C and Kalyanaraman R 2006 Nanostructure and microstructure of laser-interference-induced dynamic patterning of co on si *J. Phys. D: Appl. Phys.* **39** 5149
- [30] Rebollar E, Sanz M, Pérez S, Hernández M, Martín-Fabiani I, Rueda D R, Ezquerro T A, Domingo C and Castillejo M 2012 Gold coatings on polymer laser induced periodic surface structures: assessment as substrates for surface-enhanced raman scattering *Phys. Chem. Chem. Phys.* **14** 15699–705
- [31] Guan Y 2004 Synthesis and ordering of nanostructures using laser light *PhD Thesis* The University of Tennessee, Knoxville
- [32] Yadavali S, Khenner M and Kalyanaraman R 2013 Pulsed laser dewetting of au films: experiments and modeling of nanoscale behavior *J. Mater. Res.* **28** 1715–23
- [33] Yadavali S and Kalyanaraman R 2013 Fabricating metal nanostructures with pulsed laser dewetting self-assembly *Adv. Mater. Process.* **171** 22–26
- [34] Munroe P 2009 The application of focused ion beam microscopy in the material sciences *Mater. Charact.* **60** 2–13
- [35] Schneider C A, Rasband W S and Eliceiri K W 2012 NIH image to ImageJ: 25 years of image analysis *Nat. Methods* **9** 671–5
- [36] Sachan R, Malasi A and Duscher G 2013 Plasmon excitations in bimetallic ag nanostructures by monochromated e-beam *Microsc. Microanal.* **19** 1510
- [37] Sachan R, Malasi A, Ge J, Yadavali S, Garcia H, Krishna H, Gangopadhyay A K, Duscher G and Kalyanaraman R 2014 Ferropasmons: intense localized surface plasmons in a ferromagnetic metal *ACS Nano* at press doi:10.1021/nm5031719
- [38] Schaëffer E, Thurn-Albrecht T, Russell T P and Steiner U 2000 Electrically induced structure formation and pattern transfer *Nature* **403** 874–7
- [39] Wu N and Russel W B 2009 Micro-and nano-patterns created via electrohydrodynamic instabilities *Nano Today* **4** 180–92
- [40] Bauerle D 2011 *Laser Process. Chem.* (Berlin: Springer)
- [41] Kelley D F and Stutzman W L 1993 Array antenna pattern modeling methods that include mutual coupling effects *IEEE Trans. Antennas Propag.* **41** 1625–32
- [42] Balanis C A 2012 *Antenna Theory: Analysis and Design* (New York: Wiley)
- [43] Polyanskiy M N Refractive index database (<http://refractiveindex.info>)